

LECTURE - 18

Monday (Continue)
30/3/26

Decomposition: (of Relations)

① When a table is divided into two; one attribute should be duplicated to join them e.g: $ABCDE \rightarrow ADE, DBC$. (D is duplicated)

② Dependency Preservation: In 3NF, we'll get all FDs preserved. means FDs will be in any one of the decomposed relation (without JOIN) whereas; in BCNF, the FDs might not be preserved and we might've to join any tables to get FDs.

③ Lossless Join: Whenever we join the decomposed tables; we should always get original table instead of any less or extra info.

eg:

Student	Course	Instructor
Ali	DB	Mark
Ahmed	DB	Robin

Decomposed on "Course" X

Course	Instructor
DB	Mark
DB	Robin

Course	Student
DB	Ali
DB	Ahmed

Join

* Similarly; $\uparrow$ can't be upon "Student". It should be upon "Instructor".

* Since there is ambiguity & new info that Ahmed might've studied DB from Mark $\therefore$ this is a lossy bad join not giving original table

SOLUTION: (to avoid lossless Join)

The attribute which is duplicated & on whose basis we're joining tables SHOULD BE **KEY** IN ATLEAST ONE OF DECOMPOSED TABLES.

e.g.: In above example; Instructor is key in Instructor+Course table (as jab bhi Mark ka naam ayege; hamesha course "DB" hi hoga).

☐ Problem with BCNF: Sometimes any FD is not preserved which means we've to join any tables to ~~let~~ let FD work on it. This is necessary when we insert stuff.

e.g.: If we've above example table in Course+Instructor & Instructor+Student form; then it's in BCNF. BUT if we'll insert ~~DB + Alyan~~ in this table ^① ~~DB + Alyan~~ ^② then there will be ambiguity that Ahmed was studying DB from Robin but now we've Alyan teaching him because of new data. Therefore we need to join both ① & ② tables to see if there is any same info already existing.

Solution for BCNF: Use 3NF but that will trade off FD preservation with increase in redundancy.

Synthesis

Algorithm: Ensures that a 3NF form is dependency preserving, lossless join and attribute preserving.

Steps:

- ① Find minimal cover
 - ② For each FD, create a new relation/table with all elements of FD.
 - ③ IF NONE of the relations in ② contains a key of original relation; then add another relation which just contains the key (mostly in case of composite key).
 - ④ Remove redundant relations after ③.
- Eg: After ③, we'll have $R_1(A, B, C, D)$ & $R_2(B, D)$. Then B, D is subset of R_1 $\therefore$ we'll remove R_2 completely.

Q: $R(A, B, C, D)$ with FDs: $A \rightarrow BCD$
 $B \rightarrow D$
 $C \rightarrow D$

① Minimal Cover: $A \rightarrow BC$, $B \rightarrow D$, $C \rightarrow D$

② $R_1(A, B, C)$, $R_2(B, D)$, $R_3(C, D)$

③ IF $A+D$ was the key which isn't in any relation above then we would've made $R_4(A, D)$.

* Activity question at the end of 29th March GCR video/slides.